Remagnetization in Keweenawan rocks. Part I: conglomerates

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Received June 9, 1980
Revision accepted November 7, 1980

Conglomerate tests are used to show that Late Precambrian Keweenawan conglomerates on the Keweenaw Peninsula, Michigan and at Mamainse Point, Ontario have suffered a partial chemical remagnetization that predated tectonic tilting. At Mamainse Point the overprint was acquired during the final normal polarity stage of Keweenawan volcanism, whereas on the Keweenaw Peninsula it occurred after cessation of volcanism and has a maximum Late Freda Sandstone age. The extent of overprinting on the Keweenaw Peninsula increases westwards along a 40 km strike length in the Copper Harbor Conglomerate, and may at least in part be related to secondary mineral zonation (including native copper) in the underlying Portage Lake volcanics. It is shown that neither of the two overprinting episodes can be responsible for the reversal asymmetry observed in Keweenawan igneous rocks.

On utilise des tests sur des conglomérats pour démontrer que les conglomérats du Keweenavien datant de la fin du Précambrien sur la péninsule de Keweenaw au Michigan et à Mamainse Point en Ontario ont subi une réaimantation chimique partielle qui précède le basculement tectonique. A Mamainse Point, la surimpression a été acquise durant le stade final de polarité normale du volcanisme keweenavien alors que dans la péninsule de Keweenaw, elle s'est produite après l'arrêt du volcanisme et son âge maximum correspondrait aux couches supérieures du grès de Freda. L'intensité de la surimpression dans la péninsule de Keweenaw augmente vers l'ouest sur une distance de 40 km le long de la direction dans le conglomérat de Copper Harbor et on peut la relier au moins en partie, à la zonation minérale secondaire (comprenant le cuivre natif) dans les roches volcaniques de Portage Lake qui sont sous-jacentes. On démontre qu'aucun des deux épisodes de surimpression ne peut être responsable pour l'asymétrie de renversement observée dans les roches ignées de Keweenaw.

[Traduit par le journal]

Can. J. Earth Sci., **18**, 599–618 (1981)

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Introduction

The global Precambrian paleomagnetic data base has an uneven geographic distribution with over 50% derived from rock units of the Laurentian Shield of North America (Irving and Hastie 1975). The North American data also have an uneven geographic distribution with almost half the studies from rocks of the Keweenawan Basin; many of the latter are concentrated in the dominantly igneous Middle Keweenawan, which may have occupied a very narrow age span (perhaps as narrow as 1140–1120 Ma, according to Silver and Green 1972).

Despite the probable short duration of Keweenawan igneous activity, paleomagnetic results from igneous units contribute to much of the polar distribution recorded by rocks of the Keweenawan succession. The nature of this distribution in terms of apparent polar wander (APW) during Keweenawan time was first outlined by DuBois (1962) and subsequently more sharply defined as the Logan Loop (Robertson and Fahrig 1971) or as Hairpin 20 (Irving and Park 1972). Much of this polar trend (Fig. 1) results from the asymmetry in two apparent reversals recorded in Keweenawan igneous rocks, the younger one of which has been detected at a number of localities around Lake Superior. Two hypotheses have been examined as a possible explanation of the asymmetry.

Palmer (1970), on the basis of conglomerate tests (Graham 1949), advocated that it was due to the addition of a common secondary component to original antiparallel normal (N) and reversed (R) polarities of remanence.[1] The direction of this secondary component was obtained from the residual mean remanence obtained from lava clasts in conglomerates interbedded with and overlying lava sequences that yielded the asymmetric reversal data. In addition, the amount of secondary remanence acquired by conglomerate pebbles was found to be sufficient to have biassed the original

[1]The terms normal and reversed follow the usage of DuBois (1962) and subsequent workers in Keweenawan paleomagnetism. These terms are not used with reference to the polarity of the present earth's field as the North American APW path has large ill-defined segments between Keweenawan and Holocene times.

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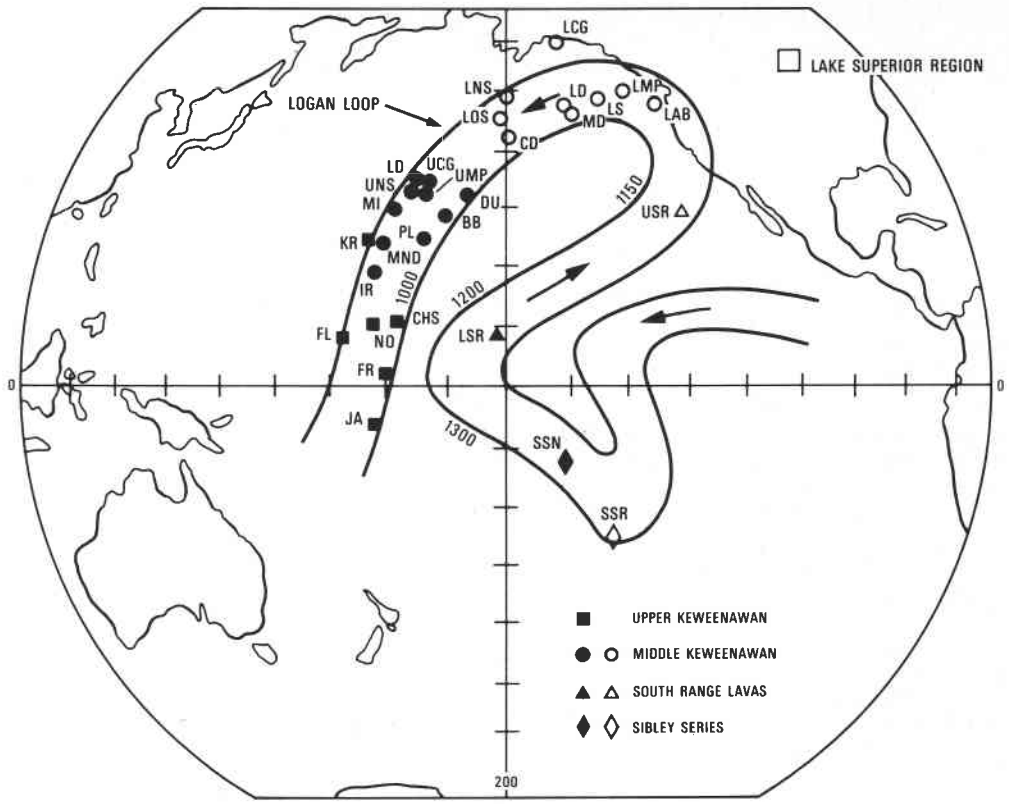


FIG. 1. Distribution of paleomagnetic poles as recorded by Keweenawan rocks of the Lake Superior region. Closed (open) symbols signify normal (reverse) polarity. Key to poles given in Table 6 of Pesonen and Halls (1979).

primary antiparallel remanence directions to their observed degree of asymmetry (Palmer 1970; Fig. 2 of Starkey and Palmer 1970). The remagnetization hypothesis (model 1) is graphically illustrated in Fig. 2*a*.

Robertson (1973) advocated that the asymmetry is the result of an exact reversal coupled with polar wander or continental drift before onset of further igneous activity in a normal field (model 2; Fig. 2*b*). Arguments in support of this hypothesis include: (a) the progressive southerly movement of the pole with time as recorded from Middle to Upper Keweenawan units in Michigan, all of which are of normal polarity; (b) Keweenawan igneous units and sedimentary rocks baked by them retain stable directions on either alternating field (AF) or thermal demagnetization, thus implying that the primary component and any overprint must have the same coercivity and blocking temperature spectra, an unlikely combination; and (c) directional data from both N and R volcanic sequences are not noticeably streaked, which requires, implausibly, that the primary and secondary components are present in the same proportion for different lava flows.

Two additional models (3 and 4 in Fig. 2) can be invoked to explain Keweenawan asymmetric reversals and may be required to reconcile evidence of repeated asymmetry in reversals at Mamainse Point (Palmer 1970; Robertson 1973; Massey 1979; Halls and Pesonen 1981). Both models appeal to departures from the standard paleomagnetic model of the Earth's time-averaged field as a geocentred axial dipole, and are based on data from the present and late Neogene fields (Wilson and Ade-Hall 1970; Cox 1975). In terms of model 3, Keweenawan reversal asymmetry could represent a magnified "far-sided pole" phenomenon mathematically equivalent to the offset dipole (Wilson 1970, 1972; Wilson and McElhinny 1974) or, in terms of model 4, a centred dipole with a secondary dipole at the core–mantle boundary (Watkins and Richardson 1974). To test the credibility of models 3 and 4 for Keweenawan poles requires synchronous data on a continental and (or) global basis. Such an assessment is currently impossible with the present limited amount of data (Halls and Pesonen 1981). Before any of the models 2, 3, or 4 are taken seriously, it is necessary to disprove the validity of model 1, the secondary magnetization hypothesis. Fortunately the effects of model 1 are entirely

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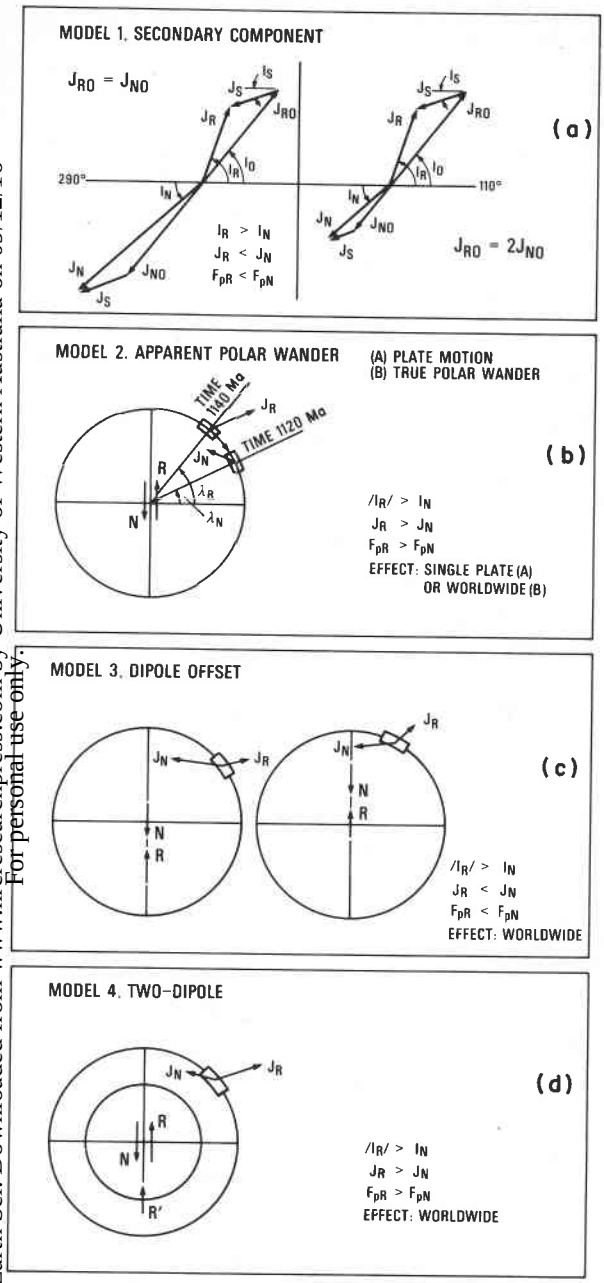


FIG. 2. Four models that can explain the pattern of Keweenawan remanence directions (from Pesonen 1978). (*a*) J_{RO} and J_{NO} are the original reversed and normal NRM intensities plotted in the observed plane of declination. J_R and J_N are resultant intensities after addition of a common secondary component J_S with westerly declination and shallow inclination. I_O, I_R, I_N, and I_S are corresponding inclinations. (*b*) One apparent polar wander model, based on plate motion. λ_R and λ_N are paleolatitudes determined from reversed and normal data. (*c*) Two versions of the dipole offset model. On the left (right) side the normal (reversed) dipole source is geocentric. (*d*) A two-dipole configuration where the major axial, geocentric dipole can have either polarity. The minor dipole R′ represents a long-term average of the nondipole field and is located at the core–mantle interface. Throughout Fig. 2, F_{pR} and F_{pN} are predicted reversed and normal paleointensities.

local; the manifestations of the process reside in the Keweenawan rocks themselves, and in theory at least, are amenable to experimental tests.

Experimental procedure

Because of the small number of measured pebbles and the ambiguities in interpretation of the earlier conglomerate tests (DuBois 1962; Palmer 1970), a more comprehensive conglomerate investigation was undertaken. Paleomagnetic sites were obtained at localities 1, 2, and 3 within the Copper Harbor Conglomerate (Figs. 3, 4) and at locality 4 within the great conglomerate of the Mamainse Point section (Figs. 3, 5). The lithology at site 4 was found to be a true conglomerate, in contrast to earlier sampling at Mamainse Point (Palmer 1970) where there was doubt as to whether the lava clasts were agglomerate fragments that may have acquired a partial thermoremanent magnetization (PTRM) on cooling (Robertson 1973). At all sites, samples were obtained only from mafic lava pebbles as these lithologically resemble Keweenawan volcanic sequences around Lake Superior that contribute most of the poles showing reversal asymmetry. All sites were from well-washed shoreline outcrops, because a surface patina and lichen cover on pebbles further inland inhibit their ready identification. In addition large shoreline outcrops were necessary because about 90% of pebbles were felsic or granitic in nature and mafic volcanic types were relatively rare (less than 10%) and tended to be of much smaller size. A total of 270 sample cores, one per pebble, were obtained using a portable field drill and oriented *in situ* using both a magnetic and a solar compass.

In the Copper Harbor Conglomerate, sites were chosen to investigate possible regional variations in magnetic overprinting along strike, and also to test whether any of this overprinting could be attributed to direct heating by interbedded lava flows (Fig. 4).

Site 1 is 525 m stratigraphically below one of these volcanic horizons, the offshore existence of which is suggested by a linear magnetic anomaly. Site 4 also lies below volcanics but in this case the stratigraphic distance is only 375 m. Site 2 has no magnetic anomaly offshore from it and thus should be free from the effects of any overlying heat sources.

Site 3 is composed of two subsites 3A and 3B, and is overlain by a sequence of about 15–20 lava flows

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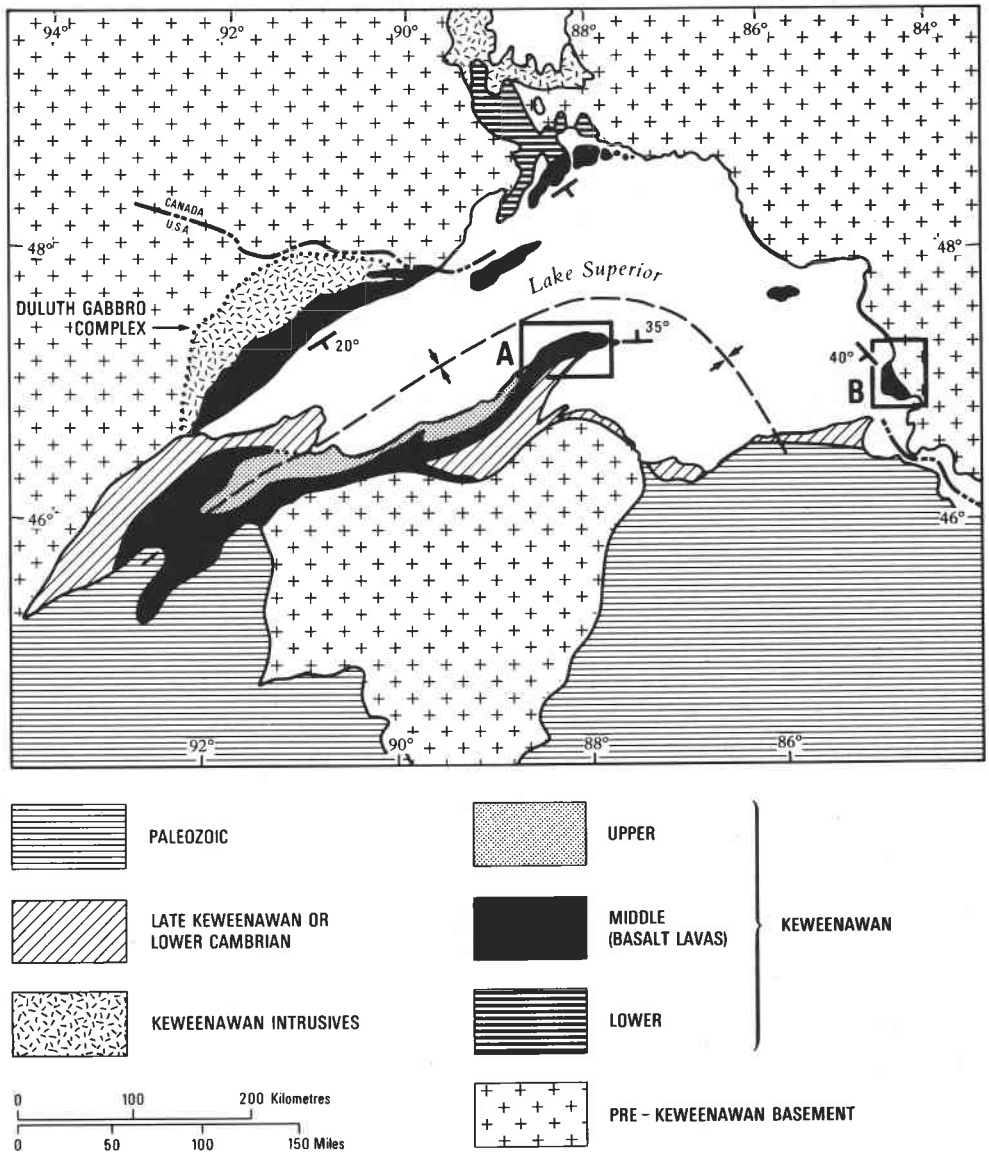


FIG. 3. Generalized geological map of the Lake Superior region, showing the location of the two study areas: A = Keweenaw Peninsula, Michigan; B = Mamainse Point, Ontario.

(Cornwall 1955). Site 3A is about 85 m below the lowermost flow. Site 3B comprises pebbles sampled from immediately beneath this flow to a distance of about 10 m stratigraphically below it. Six cores were also obtained from the flow itself. Site 3 was thus designed to gauge the extent of remanence acquisition in the conglomerate by direct lava heating and thus constitutes a baked contact test (Everitt and Clegg 1962).

In the field, sample cores were numbered consecutively and after laboratory preparation their natural remanent magnetization (NRM) was measured on a Schonstedt DSM-1 spinner magnetometer at the University of Western Ontario (UWO). Even-numbered samples were then remeasured using the Schonstedt instrument with progressive AF demagnetization in 5 mT steps to a routine maximum of 100 mT. Demagnetization was performed using a three-axis tumbler (Roy *et al.* 1973) with a solenoid in which the AF field was varied by an induction-type voltage regulator (General Electric Co. "Inductrol"). Odd-numbered samples were measured with a DIGICO spinner magnetometer at the

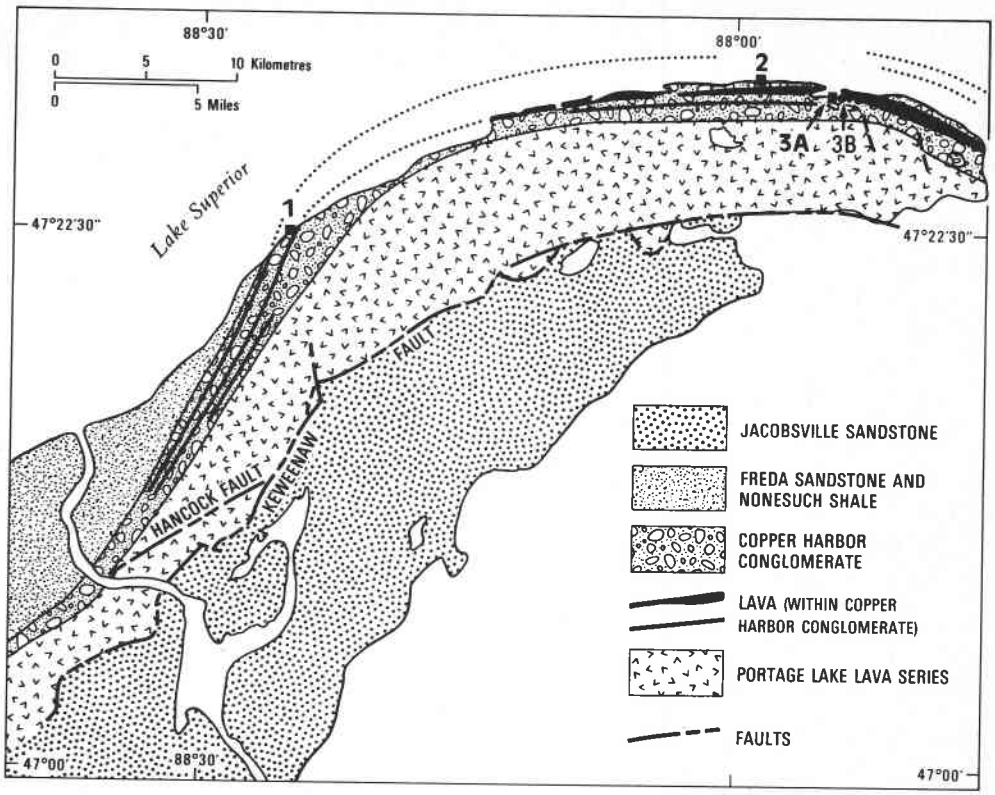


FIG. 4. Geological map of the Keweenaw Peninsula, showing location of sites 1, 2, and 3 within the Copper Harbor Conglomerate. The locations of thin lava flow units within the conglomerate are also shown and their offshore extensions by dotted lines based on aeromagnetic evidence.

University of Toronto (UT) where AF demagnetization to 100 mT was carried out using a Schonstedt GSD-1 single-axis unit.

Conglomerate test results

Distribution of NRM directions

The initial NRM directions for each of the four congomerate sites show a significant bias to westerly declinations, a bias which increases on AF demagnetization (Figs. 6, 7). This increase is due to a systematic westerly migration of about a third to one half of the pebbles at each site. From about 20 mT onwards these "swinging" specimens progressively move along great circle paths, but in only one or two pebbles is a stable end point achieved after demagnetization to maximum fields of 100 or 110 mT (Fig. 8). Figure 8 also shows that some pebbles exhibit an initial "tail" corresponding to the early removal in fields less than 20 mT of a relatively soft component with a steep downward inclination. About 60 pebbles show this effect but of these only 8 yield initial linear segments (with three or more points) on vector diagrams, thus permitting a reliable directional estimate to be made of the soft component. The mean direction obtained (345°, +79°, k = 31, α_{95} = 10°) is within 5° of the present Earth's field (PEF) direction. It is thus concluded that for most pebbles the initial tail is due to a component of recent viscous origin.

The majority of pebbles at each site are stable with directional changes of less than 2° even after intensity decay to 20% of initial value using AF fields of 100 mT. Other pebbles show similar behaviour but after early removal of the PEF component.

Statistical analysis

The results of conventional Fisher analysis applied to each site at all demagnetization levels are shown in Figs. 6 and 7 and summarized in Table 1. The significance of the nonrandom distribution in directions is demonstrated for all sites by values of R, the resultant vector, which are generally much greater than R_0, the value exceeded with a probability of 5% in sampling a population with κ = 0 (Watson 1956; Vincenz and Bruckshaw 1960; Irving 1964). For each site the value of R as a function of demagnetization field passes through a minimum at about 20–25 mT. This minimum occurs because the PEF component is first removed below 25 mT before

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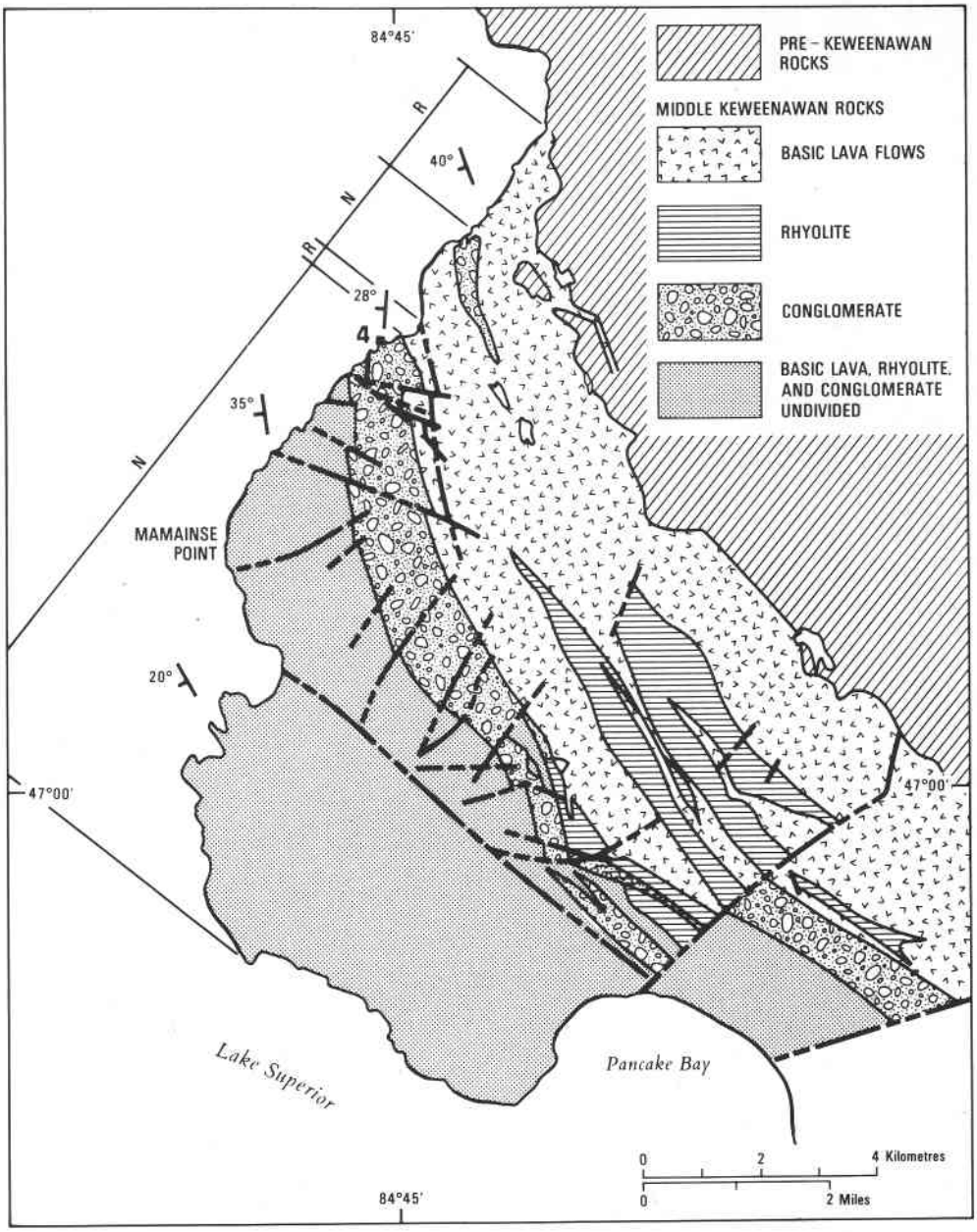


FIG. 5. Geological map of the Mamainse Point area (after Giblin 1969) showing location of site 4. Polarity divisions N, R are taken from Palmer (1970).

samples begin their westerly migration. The gradual rise in R above 25 mT corresponds to increased clustering of directions toward the west. This clustering therefore represents the effect of a secondary component of magnetization acquired after the formation of the conglomerate, which is more resistant to AF demagnetization than the random primary components carried by individual clasts. The suggestion by Robertson (1973) that the biassing arises from a preferred pebble orientation due to pebble shapes being determined by an original joint pattern is thus untenable. In any case this mechanism leads to a bimodal or girdle distribution rather than the observed unimodal one.

Due to their large values of α_{95} (20°+) the four site mean directions are not significantly different ($p = 0.05$) from one another either before or after structural correc-

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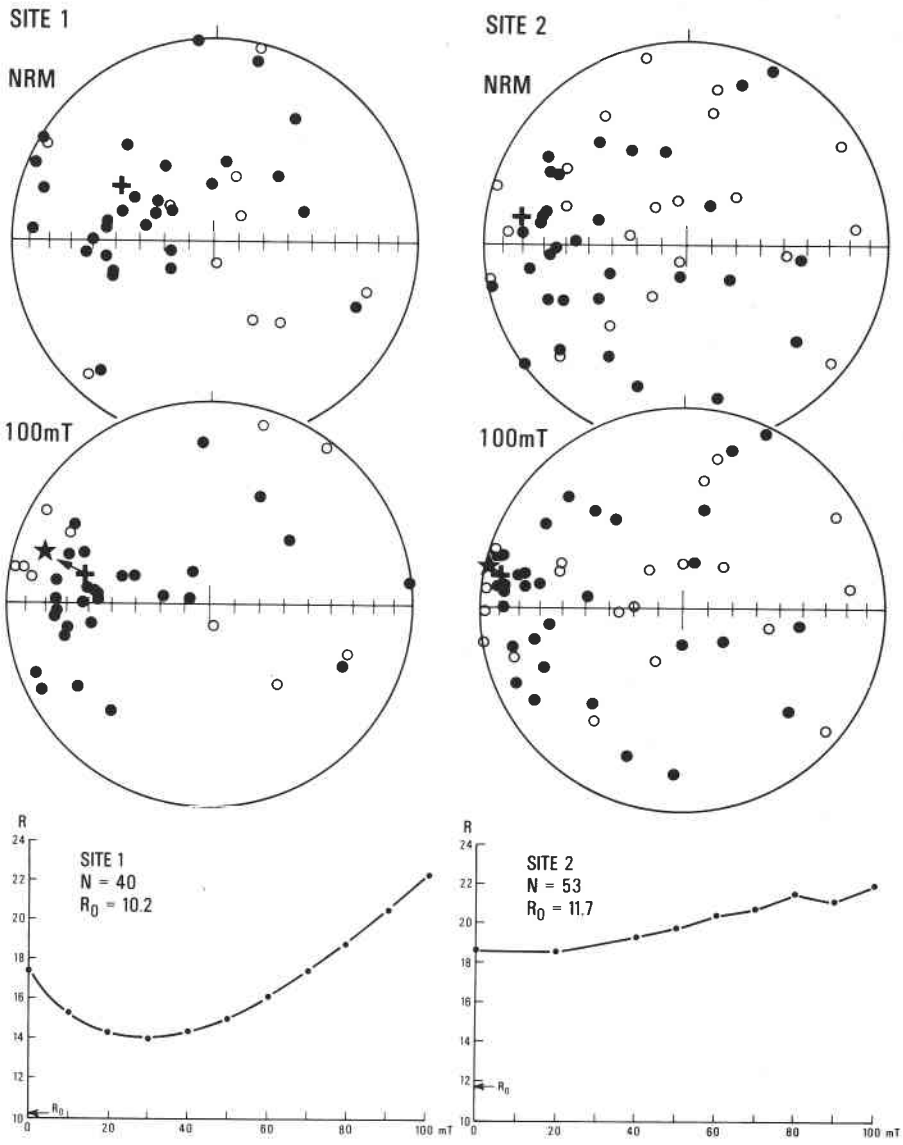


FIG. 6. Structurally uncorrected directions of magnetization in conglomerate clasts from sites 1 and 2. Closed (open) symbols represent north-seeking directions on the lower (upper) hemisphere of equal-area projections. The mean direction for each site is indicated by a cross, and after structural correction by a star. The lower part of the figure shows the variation in the length of the resultant R at the alternating fields indicated. R_0 is the length of the resultant that will be exceeded with a probability of 5% in sampling from a population with $\kappa = 0$.

tion although the precision parameter k on the mean of the four site means increases from 22.9 to 77.5 upon structural correction. This result suggests that the overprint predates tilting and may also suggest that both the Copper Harbor and Mamainse overprints were formed at the same time.

However, this latter inference is not substantiated by two further methods of analysis applied to the same data set. These methods involve the use of modes (Van Alstine 1980) and converging remagnetization circles (e.g., Halls 1976). The modal analysis tends to define local concentrations of directions in a data set that is otherwise highly dispersed. If this method is applied to the data in Figs. 6 and 7, the results agree closely with those derived from Fisher statistics, except for site 4 at Mamainse Point where the mode gives a direction having an inclination more than 20° greater than the Fisher mean. After structural correction this direction is

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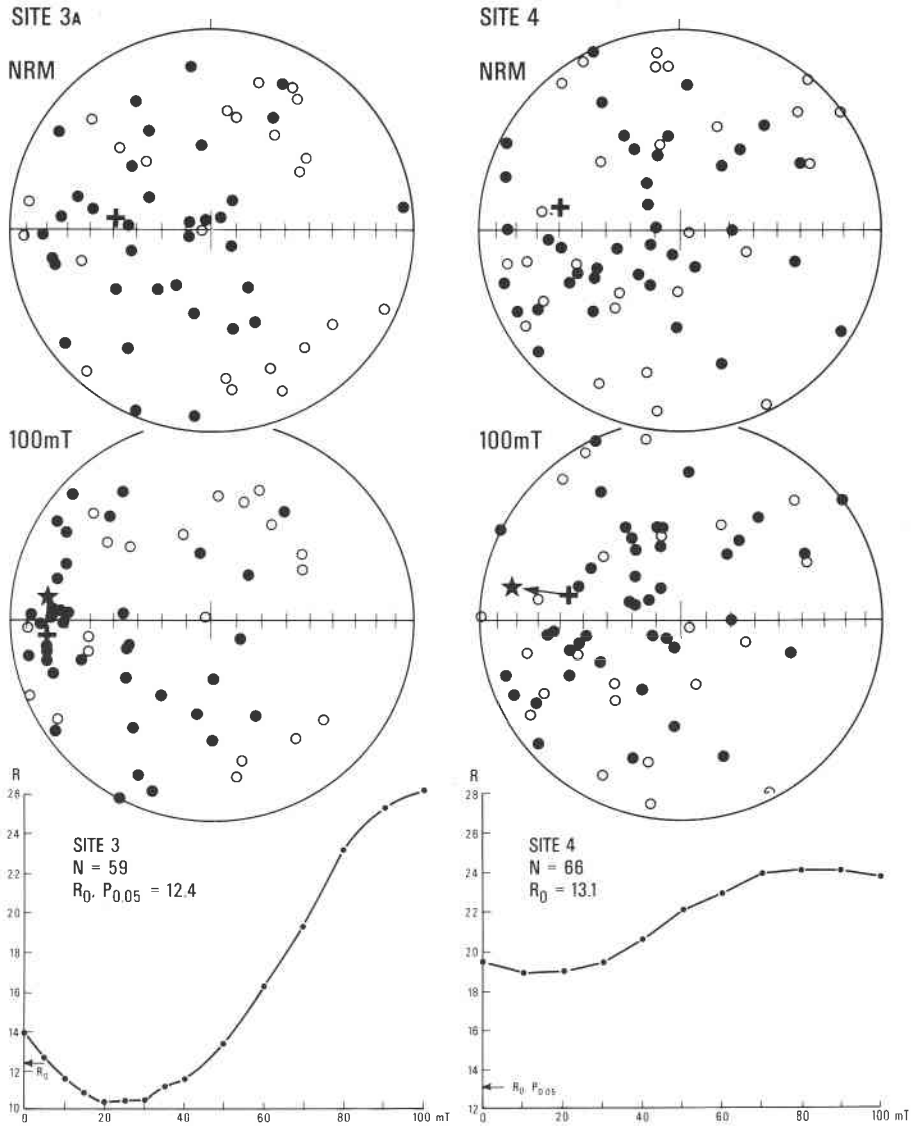


FIG. 7. Structurally uncorrected directions of magnetization in conglomerate clasts from sites 3A and 4. Symbols as in Fig. 6.

significantly different ($p = 0.05$) from those given by the three Copper Harbor sites (Table 1).

Remagnetization circles

In the remagnetization circle method, a stereonet is first used to check whether points corresponding to directions obtained after successive demagnetization steps lie along great circle paths. Great circles are then fitted to these points by least squares (Halls 1976) and circles are rejected for Q values greater than 10^{-5} (Halls 1978). For each of the four conglomerate sites, the remagnetization circles that are exhibited by many pebbles after removal of any initial PEF component show a clear convergence to a secondary component direction. The normals to each specimen great circle are thus themselves distributed along a great circle, the normal to which defines the secondary direction. Since pebble directions initially show a pronounced westerly bias the great circle normal with westerly declination is chosen rather than the antipodal easterly one.

The results of the great circle analysis (Fig. 9; Table 2) may be compared with those of the previous methods:

(1) The Copper Harbor site directions are even more tightly grouped after structural correction, compared with the results of either the Fisher or the modal analysis. Furthermore, the 95% confidence ovals about each site

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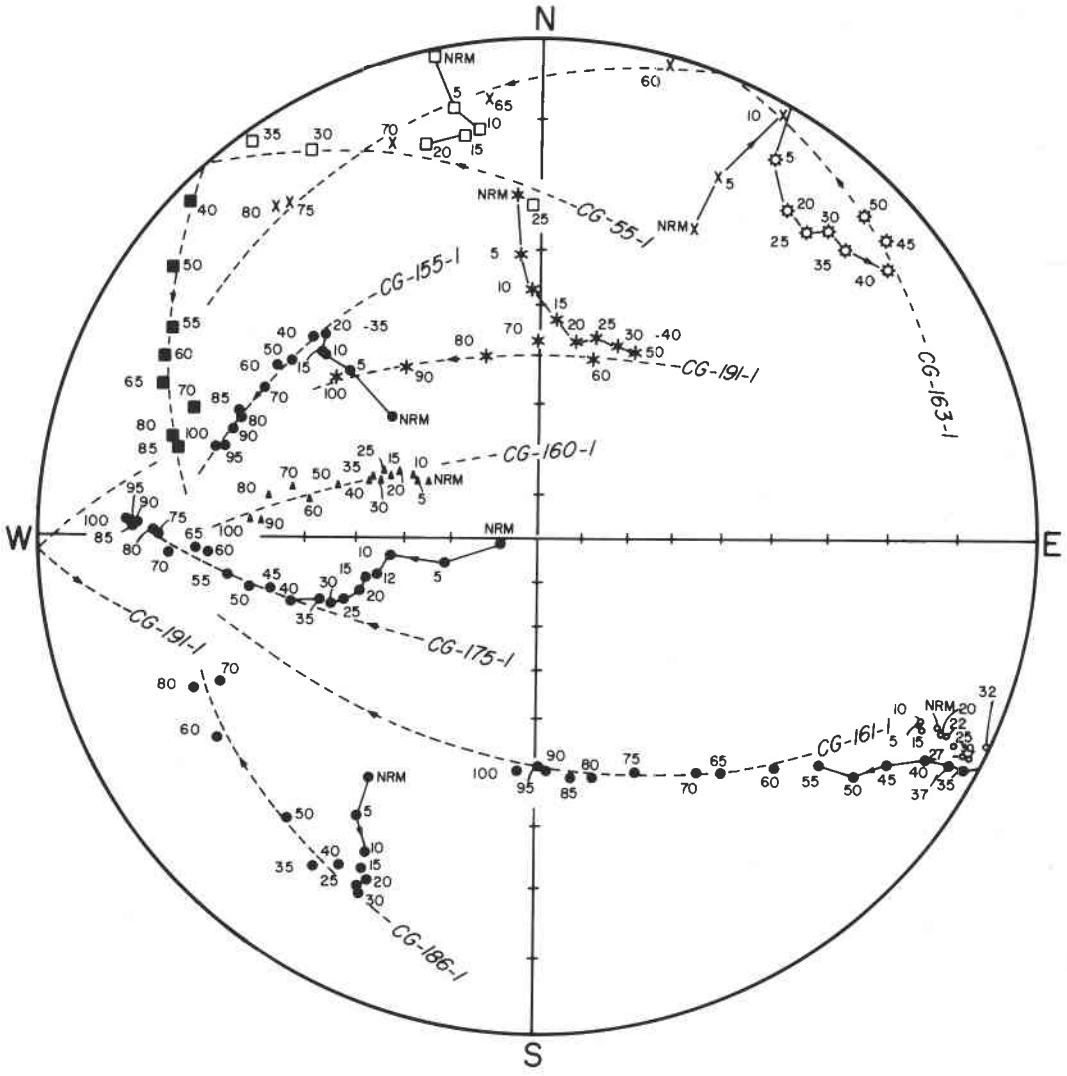


FIG. 8. Wulff net showing remagnetization circles before structural correction for a selection of clasts from sites 2 and 3A. Successive magnetization directions are shown after stepwise AF demagnetization from 0 to 80 or 100 mT. Dashed arcs are remagnetization circles fitted to those points following a great circle trend. Closed (open) symbols are as in Fig. 6 except for CG-191-1, which has all upward-directed magnetizations. Arrows indicate sense of directional movement along the demagnetization path.

direction (Fig. 9) before structural correction are such that the directions for sites 1 and 3A are significantly different ($p = 0.05$). This observation, together with the significant ($p = 0.05$) decrease in dispersion after structural correction (Table 2), indicates that the secondary component was acquired in the Copper Harbor Conglomerate *prior* to folding.

(2) In agreement with the modal analysis, the direction of the secondary component at Mamainse Point appears, after structural correction, to be significantly different ($p = 0.05$) from the Copper Harbor directions (Table 2) and resembles instead the mean primary direction obtained from the overlying lavas by Palmer (1970) and Robertson (1973).[2]

The great circle analysis thus indicates a pretilting age of the Copper Harbor secondary component (a result that

[2]In reply to a reviewer's scepticism concerning the reality of this directional difference we feel it is unlikely that units had a 10° dip *away* from Lake Superior at the time of overprinting (necessary to bring the remanence directions of Mamainse Point and the Copper Harbor into coincidence). Any pre-remanence tilt would be *towards* the lake in response to progressive subsidence of the Lake Superior Basin throughout Keweenawan time (e.g., White 1971).

TABLE 1. Mean and modal analyses of 100 mT data with structural correction

Site								
No.	Lat. (N)	Long. (W)	N	$D°$	$I°$	k	$\beta_{95°}$	$\alpha_{95°}$
1 (mean)	47.4	−88.4	40	288.4	+15.1	2.2		20½
1 (mode)	47.4	−88.4	21(33)	280.4	+17.2	17.5	7.8	
2 (mean)	47.5	−87.9	52	280.9	+01.1	1.7		22½
2 (mode)	47.5	−87.9	17(38)	279.8	+02.9	35.0	6.1	
3A (mean)	47.5	−87.9	58	273.7	+19.4	1.8		20½
3A (mode)	47.5	−87.9	16(44)	280.1	+22.2	45.1	5.5	
1 +2 + 3A			150*	280.6	+12.3	1.9		12
4 (mean)	47.1	−84.7	66*	280.8	+14.5	1.5		23½
4 (mode)	47.1	−84.7	15(49)	297.4	+39.9	19.9	8.8	

NOTES: For the Fisher mean analyses N is the number of pebble directions used whereas for the modal analyses it is a subset of the bracketed number (which defines the mode) that has a Fisher mean identical to the mode. β_{95} is the α_{95} of this subset (Van Alstine 1980). $D°$, $I°$, and k are the declination, inclination, and precision parameter, respectively.

*Although 219 unbaked pebbles were collected, 3 were thermally demagnetized without prior AF demagnetization.

was less secure using the other methods), and shows it to have a similar direction to that of the primary component from the overlying Nonesuch Shale (279.8°, +9.8°) reported by Henry *et al.* (1977). Furthermore, in contrast to the Fisher analysis, both the great circle and modal analyses suggest that the Mamainse overprint was acquired at a different time. From consideration of the apparent polar wander path (Fig. 1), the Mamainse overprint may have formed during Middle Keweenawan volcanism and thus preceded the Copper Harbor overprint.

The effect of interbedded flows on conglomerate NRM

A possible cause of the partial overprint found in the Copper Harbor Conglomerate is that it is due to heating by interbedded lava flows, or due to the effects of a raised geothermal gradient during times of Copper Harbor volcanism. This interpretation can be tested from results obtained at site 3B, which are summarized in Fig. 10.

Five of the six cores from the 10 m thick lava flow were progressively AF demagnetized whereas the sixth was subjected only to thermal treatment. Both demagnetization procedures indicate single component behaviour and thus a stable remanence direction. Pebbles adjacent to the flow contact (within 1–2 cm) when subjected to AF and thermal demagnetization also show the same result both in degree of stability and in the remanence direction (Pesonen 1978). The same component also completely dominates the remanence in pebbles further from the contact out to a distance of 1 m, and is present in clasts 1–5 m from the contact (Fig. 10). Pebbles sampled at distances greater than 5 m are more dispersed although there is a slight westerly bias; this bias may reflect the regional overprint as seen in site 3A. A fully positive baked contact test can thus be demonstrated; the lava flow has remagnetized or partially remagnetized only those pebbles within about 5 m of the contact. The single component of magnetization found in the flow suggests, in the light of the positive baked contact test, that it is a primary component acquired during initial laval cooling. On correction to the paleohorizontal the remanence direction (D = 292°; I = +38°; k = 272) is very close to that of the underlying Portage Lake volcanics (Books 1972).

A detailed investigation of seven further sites within Copper Harbor volcanic units (Halls and Palmer 1981), distributed along a strike length of about 40 km, demonstrates that the same primary component is present as that found at site 3B. Since the inclination of the conglomerate overprint found at sites 1, 2, and 3A is more than 20° less than that given by the intercalated volcanics, it may be concluded that the overprint is *not* due to direct heating by flows or to increased thermal gradients or thermochemical activity during episodes of Copper Harbor volcanism. It is an overprint that postdates the formation of the entire Copper Harbor sequence. Further support for this conclusion is that certain sites within the volcanic units have been affected by this overprint, but discussion of these results will be reserved until later (Halls and Palmer 1981).

Nature of conglomerate overprints

Thermal demagnetization

Thermal demagnetization on about 20 selected samples was performed both at UWO (Schonstedt TSD-1 unit) and at UT (PAR spinner, and furnace constructed by LJP). For the majority of samples, thermal demagnetization followed AF treatment, whereas for the remainder, second specimens of core samples were used that had not previously been demagnetized.

For approximately half the specimens so examined,

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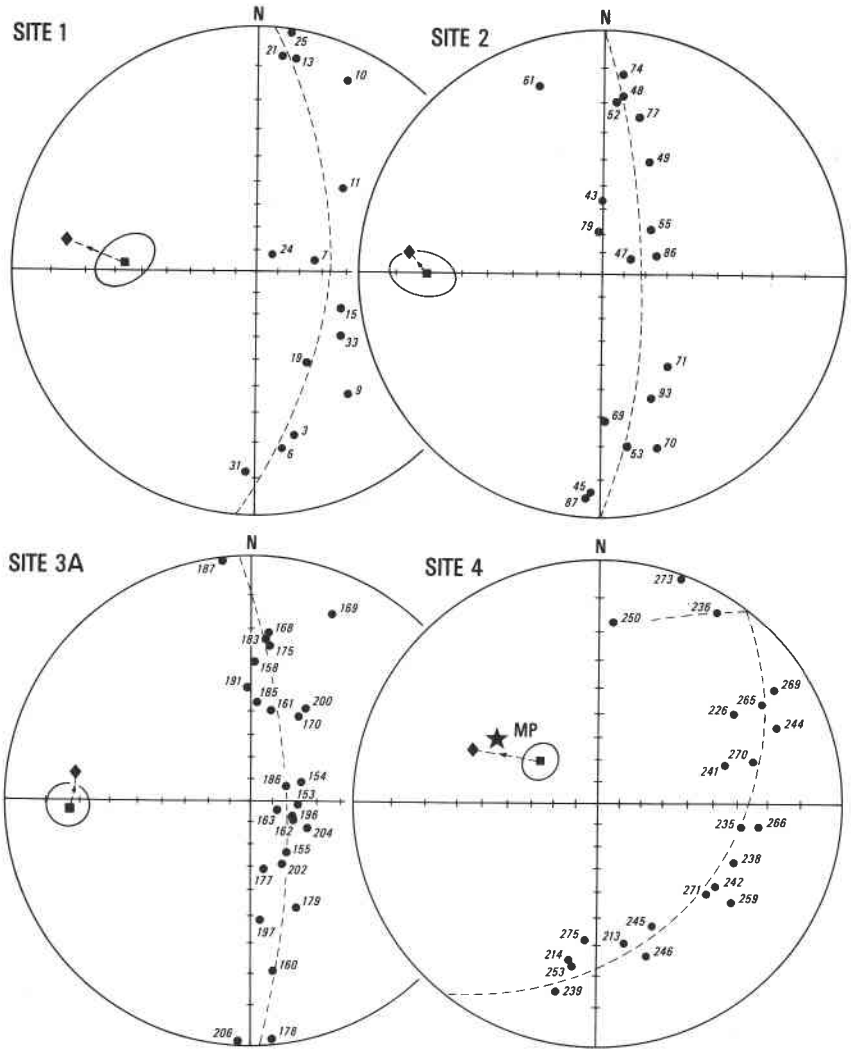


FIG. 9. Pebble remagnetization circle data for each conglomerate site. Solid circles are downward-pointing normals to pebble remagnetization circles; solid square is the downward-pointing normal to the great circle (dashed line) fitted by least squares to the pebble normals, and gives the direction, before structural correction, of the secondary component. Ovals of 95% confidence drawn about each direction are specified in Table 2. The solid diamond is the direction of the secondary component after restoring conglomerates to the paleohorizontal; the direction at site 4 is within 10° of that given by Mamainse Point volcanics of normal polarity and shown by a star.

the apparent end point obtained by AF demagnetization was confirmed by the thermal treatment up to temperatures in excess of 600°C. In others, great circle swings observed with AF demagnetization to 100 mT were extended by subsequent thermal demagnetization (Fig. 11). Since in most cases the great circle swings are toward the direction of the secondary component (Fig. 11 *a*, *b*, *d*), this secondary component is not merely harder, but has an unblocking temperature appropriate to that of hematite or titanohematite. That is not to say, however, that the primary component is exclusively carried by magnetite and the secondary component by hematite. Some of the hematite in Keweenawan lavas is the result of primary high-temperature deuteric oxidation (Ade-Hall *et al.* 1971). A few samples exhibit swings away from the secondary component direction with AF demagnetization indicating a primary component harder than the secondary component. One such example is given in Fig. 11*c* where the harder component has unblocking temperatures inappropriate for magnetite. In this particular example, there is evidence at the 625 and 640°C steps for a reversal in movement along the great circle, suggesting the presence of both primary and secondary hematite having slightly different unblocking temperatures.

The thermal results show the overprint to be a

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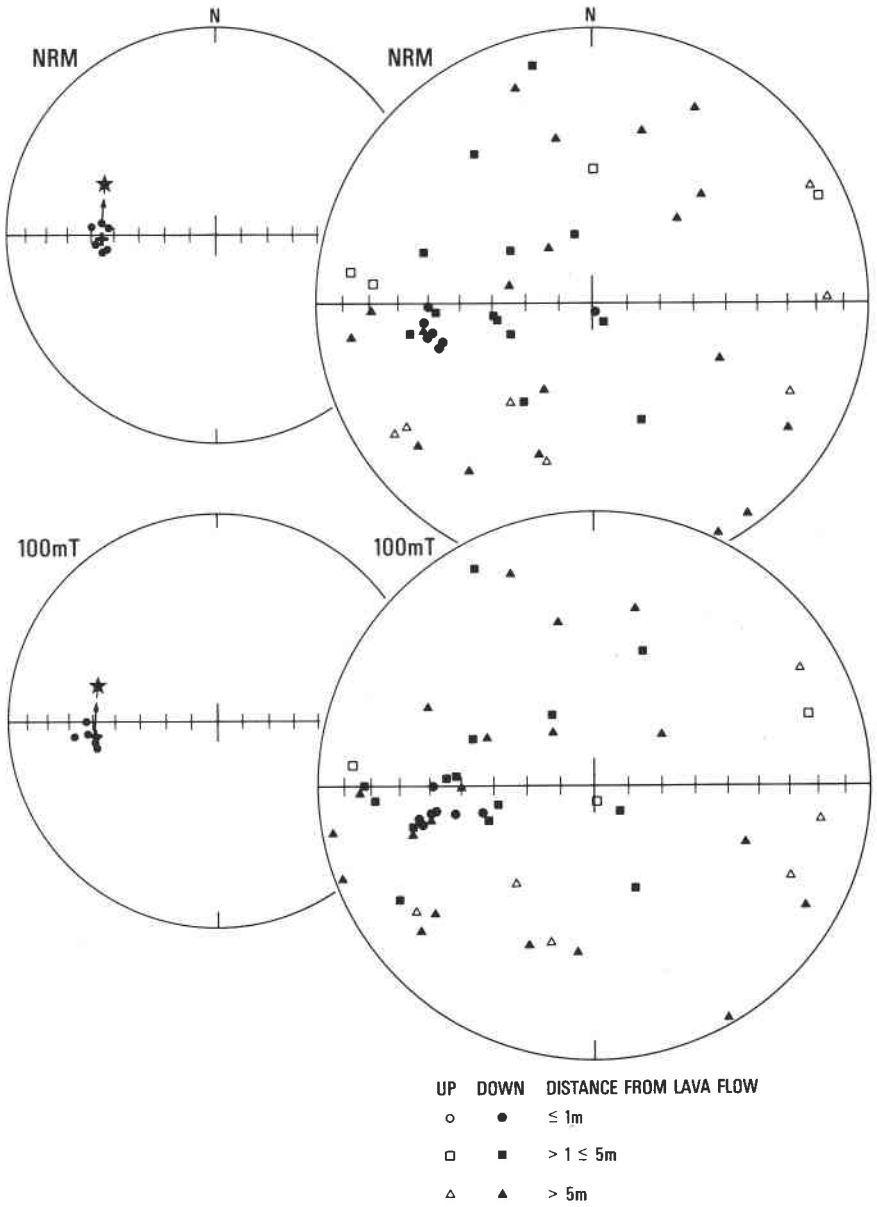


FIG. 10. The left part of the figure shows individual sample magnetizations from the lava at site 3B. The lava mean is indicated by a cross that, when structurally corrected, moves to the position of the star. The right part of the figure indicates the directions in conglomerate clasts collected at varying stratigraphic distances from the underside of the lava flow. Note that all pebbles collected within 1 m of the lava possess remanent magnetizations similar to that of the lava flow whereas samples collected away from the baked zone (triangles) are widely dispersed.

chemical remanent magnetization (CRM) since this component is erased only at the highest blocking temperatures. Had it been a TRM, for example, all magnetic grains would have been reset and swinging behaviour on demagnetization would not have been observed.

AF demagnetization and intensity relationships

For each of the four conglomerate sites a series of histograms is presented in Fig. 12; they show the distribution of sample directions after a demagnetization field of 40 mT as a function of angular distance β from the site secondary overprint direction. In each case the sample population is divided into relatively low- and high-intensity groups and these are plotted separately. A further distinction is also made in Fig. 12 concerning the amount of directional swing θ during the demagnetization interval between the step at which the PEF

TABLE 2. Remagnetization circle analysis

Site No.	Lat. (N)	Long. (W)	N_{RC}	$D°$	$I°$	$E°_{min}$	$E°_{max}$	$D°_{min}$	$I°_{min}$	$D°_{max}$	$I°_{max}$
Copper Harbor Conglomerate, Michigan (no SC)											
1	47.4	−88.4	14	274.3	33.3	10	13	170.1	20.6	54.1	49.4
2	47.5	−87.9	18	267.2	17.8	8.3	11	1.2	12.1	303.8	−68.2
3A	47.5	−87.9	26	269.1	17.4	5	5.6	29	57.8	350.2	−26.2
Great Conglomerate, Mamainse Point, Ontario (no SC)											
4	47.1	−84.7	22	300.8	58.4	7.2	7.4	313.4	−30.9	40.0	5.6

	No SC						With SC			
Site	$D°$	$I°$	$D_1°$	$I_1°$	ω°	δ°	$D°$	$I°$	$D_1°$	$I_1°$
1	274.3	33.3	275.0	32.7	225	25	280.9	13.3	281.2	12.7
2	267.2	17.8	267.2	17.8	265	35	277.2	13.3	277.2	13.3
3A	269.1	17.4	269.1	17.4	275	28	278.2	18.1	278.2	18.1
4	300.8	58.4	298.8	60.6	185	28	290.6	32.0	288.4	35.2

SC	N_S	$\bar{D}_1°$	$\bar{I}_1°$	k	α_{95}	P lat. (N)	P long. (W)	δM	δP
Copper Harbor Conglomerate (pole positions)									
No	3	270.2	22.7	73.5	14.5	8.8	170.0	15.4	8.1
Yes	3	278.9	14.7	511	5.5	11.5	180.7	5.6	2.9

NOTES: N_S is the number of sites; N_{RC} is the number of pebbles yielding remagnetization circles; D, I are site declination and inclination; D_1, I_1 are same corrected to a common site (lat. 47.5°N, long. 87.9°W); k, α_{95} are usual Fisher statistical parameters; E_{min}, E_{max} are the 95% confidence errors in the directions (D_{min}, I_{min}) and (D_{max}, I_{max}), respectively, given by error analysis program of Bailey and Halls (1978); ω, δ are the observed strike and dip of conglomerate bedding; P lat. and P long. are latitude and longitude of paleopole, respectively; δM, δP are the semi-axes of the 95% confidence oval about the pole; and SC is structural correction.

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component is finally removed (~20–25 mT) and the maximum step (usually 100 mT). Samples are classified as stable if $\theta \leq 5°$ and "swinging" if $\theta > 10°$.

The AF demagnetization results of all four conglomerate sites shown in Figs. 6 and 7 are summarized in Table 3. It is seen that swinging samples are generally softer to AF demagnetization and of weaker intensity, and that after demagnetization to 40 mT their directions tend to lie within about 50° of the overprint direction. There *are* some very weak, *stable* samples but most of these have directions within 30° of the overprint direction and are interpreted to have been almost totally reset. In summary, therefore, the intensity relations shown in Fig. 12 show that relatively weak samples swing more on AF demagnetization and yield directions that lie closer to that of the overprint, compared with stronger samples. The latter tend to yield randomly distributed directions that remain invariant to AF demagnetization.

The above relations, like the thermal demagnetization data, also indicate that the overprint must be a CRM, the formation of which has in some way involved the destruction of magnetite and the appearance of a weaker but harder magnetic phase such as hematite. If the overprint were a pure PTRM then the more strongly magnetized samples would tend to yield directions closer to that of the secondary component.

Figure 13 shows β histograms for each site where samples are separated according to the interpreted presence or absence of CRM. These histograms include all samples regardless of intensity and are modified from those in Fig. 12 by transferring any weak totally reset samples from left- to right-hand side histograms.

For sites 1, 2, and 3A, the histograms are generally flat for samples "with no CRM" thus indicating random directions, whereas for site 4 at Mamainse Point there is a definite bias toward low β values in samples supposedly uncontaminated by CRM on the basis of their stability. The reason for this behaviour (and also the general lack of θ values above 5°) is that of all the conglomerate samples those at site 4 were generally more resistant to AF demagnetization (Table 3). It might be suspected, therefore, that failure of most Mamainse samples to swing is more a consequence of hard two-component remanence than of single-component behaviour. This conclusion was verified in part as a number of samples showing very small swings ($\theta < 2°$)

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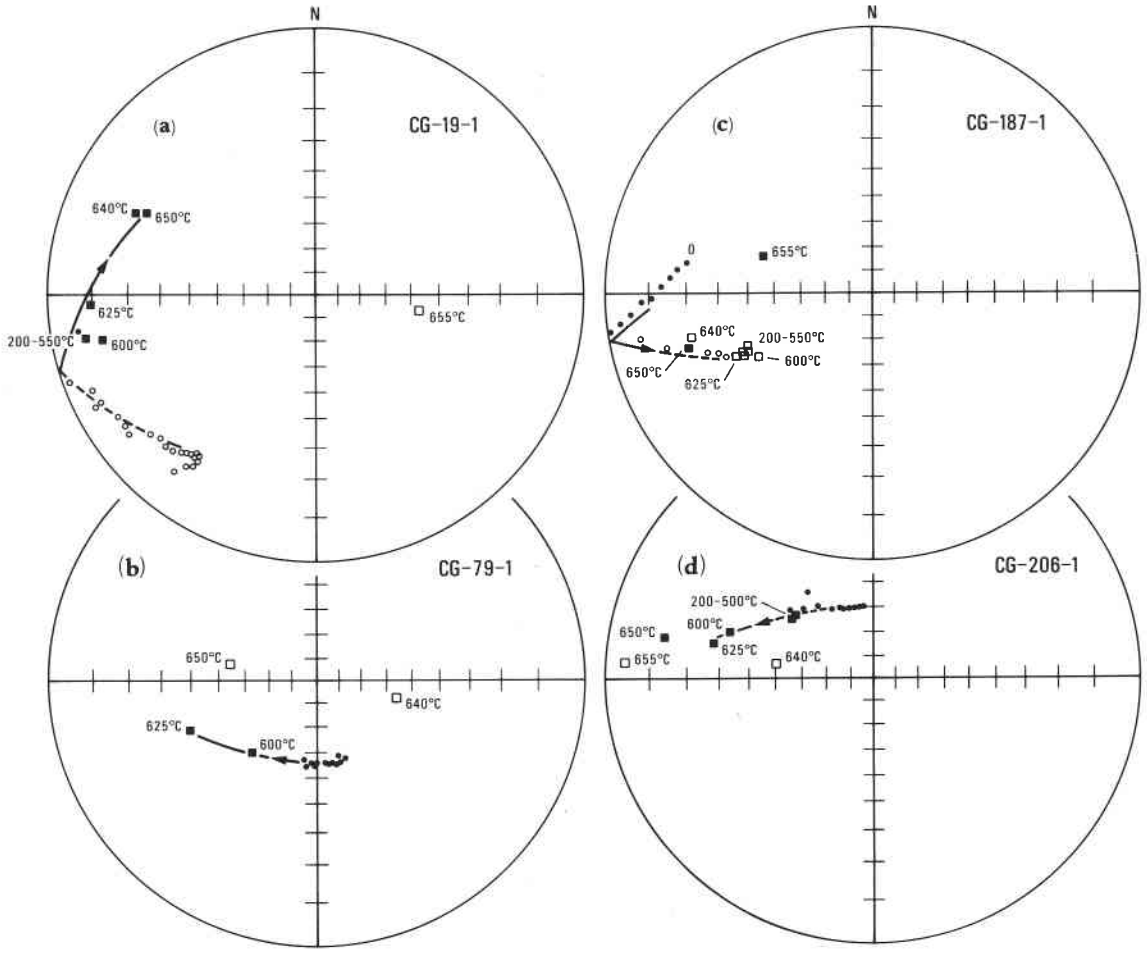


FIG. 11. Wulff nets showing pebble remagnetization circles obtained by AF demagnetization (small circles) and thermal demagnetization (squares). Closed (open) symbols represent north-seeking directions on the lower (upper) hemisphere of the projections after structural correction. CG-19 is from site 1; CG-79 is from site 2; and CG-187 and CG-206 are from site 3A.

after AF treatment exhibited more pronounced swinging behaviour after thermal demagnetization (Pesonen 1978). The "thermal" swings were generally in the opposite sense to those produced by AF means and were directed away from the overprint direction.

The relative difficulty in component separation encountered at site 4 does not appear to be caused by a lower degree of resetting than at Copper Harbor sites. In fact the degree of directional bias for site 4 (obtained from the ratio R/N in Fig. 7) is similar to that at site 2 in Fig. 6. However, the Mamainse samples are generally much redder than those at sites 1, 2, and 3A (see next section) and their intensity is generally less than half (Table 3). These observations may indicate that prior to overprint acquisition Mamainse samples already contained more hematite, which would inhibit detection through AF demagnetization of any CRM formed by later hematite growth.

Returning to the Copper Harbor Conglomerate, an important aspect concerns the relative level of overprinting between the three sites 1, 2, and 3A. Both Figs. 6 and 13 suggest a greater extent of CRM overprinting in the most westerly site 1 than in the other two. There is no indication either from petrographic examination or from Table 3 that the pebble population at site 1 was originally any different from those at sites 2 and 3, so it is surmised that the greater degree of resetting at site 1 reflects a greater intensity or duration of the remagnetization process. Further support for this conclusion is that zeolites and other secondary minerals in the underlying Portage Lake volcanics (PLV) indicate that temperatures and the activity of chemical solutions also increase along strike to the west (Stoiber and Davidson 1959). Whether these effects are related to burial depth or reflect local variations in geothermal gradient is presently unknown. One puzzling aspect is that if the

TABLE 3. Remanent magnetization statistics for conglomerate pebbles

Stability	N	J_0 (× 0.1 A/m)	J_{40} (× 0.1 A/m)	J_{40}/J_0
		Site 1		
Stable ($\theta < 5°$) (1 rejected, $J_0 = 100$)[1]	11	6.10±3.45	4.61±3.05	
Intermediate ($5° < \theta < 10°$)[2]	7	4.57±5.9	2.53±3.9	
Unstable ($\theta > 10°$)[3]	21	2.86±1.07	1.06±0.33	
No CRM[a]	7	8.77±4.73	6.67±4.22	0.71±0.22
With CRM[b]	32	3.05±1.25	1.31±0.73	0.46±0.07
		Site 2		
Stable[1]	28	5.72±1.79	4.79±1.67	
Intermediate[2]	7	2.14±2.06	1.68±1.77	
Unstable[3]	18	0.95±0.54	0.66±0.34	
Minor CRM[c]	25	6.37±1.91	5.34±1.79	0.83±0.06
Major CRM[d]	28	1.17±0.56	0.87±0.44	0.73±0.07
		Site 3A		
Stable[1]	19	7.44±1.91	5.02±1.35	
Intermediate[2]	11	4.61±2.12	2.53±1.34	
Unstable[3]	28	2.90±1.18	1.22±0.55	
No CRM[e]	19	7.44±1.91	5.02±1.35	0.77±0.08
With CRM[f]	39	3.39±1.0	1.59±0.53	0.53±0.06
		Site 4		
Stable ($\theta < 2°$)[1]	38	2.98±0.69	2.65±0.60	
Intermediate ($2° < \theta < 5°$)[2]	14	2.99±1.7	1.80±0.91	
Unstable ($\theta > 5°$)[3]	13	0.48±0.17	0.28±0.06	
Minor CRM[g]	50	3.08±0.66	2.50±0.60	0.86±0.03
Major CRM[h]	15	0.49±0.14	0.30±0.06	0.69±0.08

NOTES: θ is the amount of NRM directional movement in degrees on AF demagnetization from ~20 to 100 mT; N is the number of pebbles; J_0 and J_{40} are the initial NRM intensity and the intensity after demagnetization to 40 mT; quoted uncertainties are 95% confidence levels.

[a]Sum of number of samples in (1) minus 4 totally reset samples (R).
[b]Sum of (2), (3), and 4R.
[c](1) minus 3R.
[d]Sum of (2), (3), and 3R.
[e]Same as (1).
[f]Sum of (2) and (3).
[g]Sum of (1) and (2) minus 2R.
[h]Sum of (3) and 2R.

remagnetization seen in the Copper Harbor Conglomerate is due to the same process that emplaced the secondary minerals in the underlying volcanics, then an overprint with shallow inclination should also occur in the volcanics. However, paleomagnetic data from the PLV show no evidence (e.g., directional streaking within site populations) that such a component is present (Books 1972).

One further point concerns the provenance of the pebbles found in the Copper Harbor and Mamainse conglomerates. The mean value of NRM intensity J_0 for stable samples from sites 1, 2, and 3A combined (about 5–7 × 10^{-1} A/m in Table 3) is very similar to that in the underlying PLV (about 10 × 10^{-1} A/m: Books 1972). However, the conspicuous absence of ophitic basalt pebbles in contrast to the frequency of this lava type in the PLV (see next section) suggests that the Copper Harbor Conglomerate may have been derived largely from an older more distant volcanic terrain, perhaps represented by the Powdermill Group[3] (Hubbard 1975). However, the intensity J_0 of the stable pebbles is much lower than that of the Powdermill Group (~30 × 10^{-1} A/m: K. G. Books, unpublished data), perhaps as a consequence of weathering during erosion at the source and subsequent pebble transportation. A similar conclu-

[3]These rocks were informally known before as the South Range traps.

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sion also holds for Mamainse Point as the mean value of J_0 for the stable pebbles (3×10^{-1} A/m) is also much less than that of the *in situ* volcanics from which they are presumably derived ($\sim 12 \times 10^{-1}$ A/m: Palmer 1970). However, since these stable pebbles may contain significant CRM acquired after conglomerate formation, it is less clear at what stage in their history the pebbles underwent their main phase of alteration.

Megascopic and microscopic petrography

The preceding thermal and AF results suggest that the secondary component was acquired through hematite growth at the expense of a more magnetic phase with a lower blocking temperature, such as magnetite. Since the colour of pebbles may thus reflect their extent of CRM overprinting, all unbaked pebbles were colour-coded using Munsell soil color charts. The resultant colour codes occupied a spectrum from red (corresponding to fine-grained hematite) through browns to greys. However, a plot of colour code versus either J_0 or β for all sites failed to show any correlation, the only positive result being that site 4 (Mamainse Point) was characterized by a greater number of red samples than the other sites.

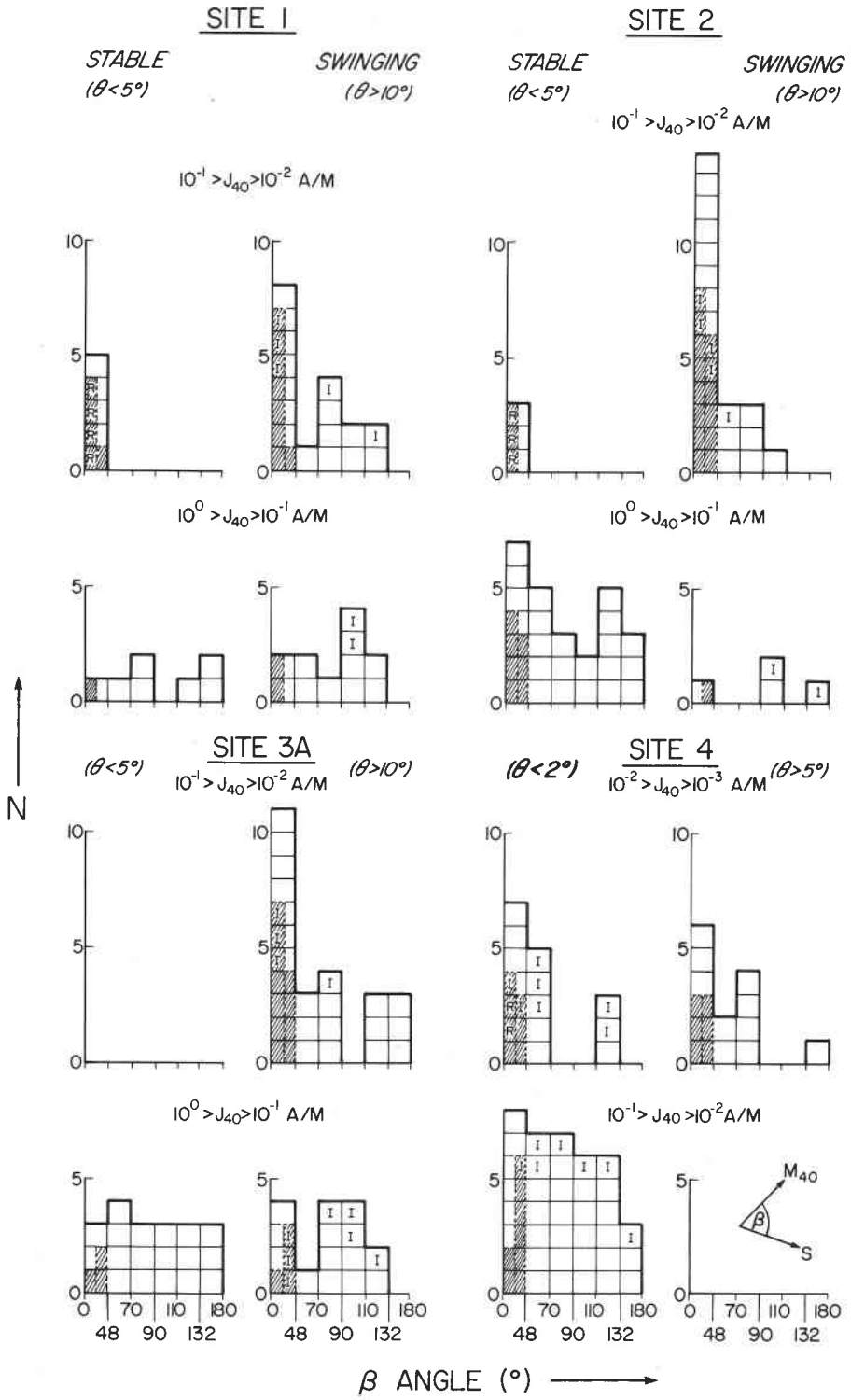
An examination of 12 polished thin sections taken from equal numbers of stable and swinging Copper Harbor pebbles suggests a plausible reason for the lack of correlation. Using reflected light and oil immersion lenses with nominal magnifications up to 1000 times, all samples contain grains that morphologically are after titanomagnetites with an original high-temperature oxidation typical of types C1 to C3 as defined and illustrated by Haggerty (1976). However most of these grains are now partially or totally hematite; original oxyexsolution lamellae of ilmenite are now black sphene as described by Ade-Hall *et al.* (1971). By far the most abundant oxide is hematite, some of which is after olivine but much appears to have resulted from the crystallization of original interstitial glassy material. Some hematite occurs in anastomosing microveinlets and is a by-product of metamorphic reactions among the silicate minerals. Because of the varying size of hematite particles in these rocks, the hematite ranges from white, highly reflective, anisotropic grains or polygranular aggregates to submicron size particles exhibiting internal reflections in red. Hematite replacing the larger original titanomagnetites is dominantly of the specularite type, which contributes little to overall reddening of the rock.

That much of the hematite in Keweenawan lavas is primary, resulting from high-temperature deuteric oxidation, has been persuasively argued by Butler and Burbank (1929). Some of the hematite was produced by burial metamorphism, but since the silicate mineral assemblages (see below) were probably produced in the parental laval flows before the subsequent erosion that provided the pebbles, the problem is one of identifying a morphologically distinct, tertiary generation of hematite in the CRM-bearing pebbles. None could be identified.

As viewed with transmitted light, all the sections examined are fine-grained altered basalts possessing intersertal or trachytic textures; most are amygdalar. Ophitic texture basalts are not present and none were noted in the megascopic examination of all unbaked pebbles. Half the samples are microporphyritic in plagioclase; a few have pseudomorphed relicts of olivine. Where porphyritic, the microphenocrysts are set in a groundmass of plagioclase, oxides, and the products of devitrified glass. Secondary minerals including chlorite, epidote, calcite, quartz, and sphene occur in both groundmass and amygdules, and represent the products of burial metamorphism. No clinopyroxene remains in any of the pebbles examined.

Relict clinopyroxene occurs throughout the upper two

FIG. 12. Histograms for each conglomerate site of the number of pebbles N as a function of β, where β is the angle in degrees between the secondary component direction S as determined from converging remagnetization circles and the direction of magnetization M_{40} after AF demagnetization at 40 mT. The β scale is such that each interval defines latitudinal bands of equal area around a sphere whose axis is parallel to S. A population of randomly directed vectors should be distributed evenly in number between the six bands, and thus a flat histogram should be obtained. For each site the histograms are separated according to (1) the directional swing in degrees θ occurring in the interval defined by the maximum demagnetization field used (normally 100 mT) and the value that is just sufficient to erase any PEF component (about 20–25 mT), and (2) the magnetization intensity J_{40} remaining after a demagnetization field of 40 mT. For sites 1, 2, and 3A from the Copper Harbor conglomerate, samples are separated according to whether they have values of J_{40} greater or less than 10^{-1} A/m, and whether they are stable ($\theta < 5°$) or swinging ($\theta > 10°$). Samples of intermediate stability ($5° \leq \theta \leq 10°$), marked with an I, have been included for convenience with the swinging group. For site 4 from Mamainse Point (which is weaker and harder to AF demagnetization), samples are divided according to whether they have values of J_{40} greater or less than 10^{-2} A/m, and whether they are stable ($\theta < 2°$) or swinging ($\theta > 5°$). Intermediate (I) samples ($2° \leq \theta \leq 5°$) are included with the stable population to retain conformity with Copper Harbor histograms. The β band defined by angles $0 \rightarrow 48$ has been subdivided to emphasize those samples with $\beta < 30°$ (the shaded histograms). Most stable samples of low intensity J_{40} that fall in this lowest β range have values of $\beta < 20°$ and are interpreted to have been totally reset magnetically. Such samples are labelled R in the histograms.



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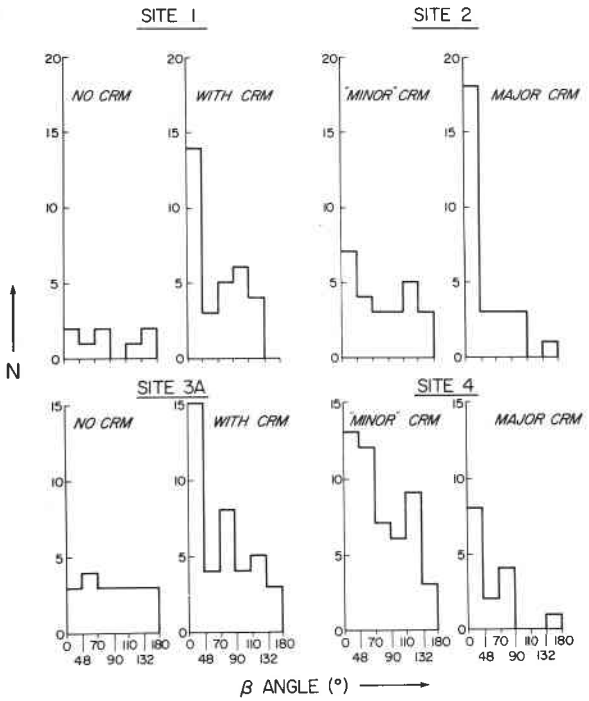


FIG. 13. β histograms of the same samples as in Fig. 12 but subdivided according to their relative degree of CRM content inferred from the paleomagnetic data. The histograms have been derived from those in Fig. 12 by (1) combining, for stable and swinging categories, samples regardless of their intensity, and (2) transferring all totally reset samples (R) from left- to right-hand side histograms.

thirds of the underlying Portage Lake volcanics but lavas possessing the metamorphic mineralogy of the pebbles occur only in the lower third of the PLV sequence (Jolly and Smith 1972). Since the level of burial metamorphism in the pebbles is higher than in the immediately underlying flows, it is likely inherited from the source region and thus predates conglomerate formation. A secondary clinoamphibole, probably actinolite, was found in two of the pebbles. Although unreported in the PLV, this mineral does occur in the older Powdermill Group (Halls 1969) and indicates a still higher level of burial metamorphism.

Secondary mineral assemblages therefore suggest that the source rocks for the Copper Harbor pebbles were the older deeper flows of the Keweenawan volcanic pile—the Powdermill Group and the lower parts of the PLV. Reservations concerning the former as a *major* source of material remain, however, because of the NRM intensity differences discussed in the previous section.

Discussion

The foregoing results demonstrate that Keweenawan conglomerates have suffered a partial chemical remagnetization. The direction of this overprint, at least for the Copper Harbor Conglomerate, is broadly that required to explain reversal asymmetry by Palmer's (1970) remagnetization hypothesis. However, there are at least three observations that strongly argue against this idea.

(1) Baked contact tests for Keweenawan normal and reversed dikes showing stable, asymmetric directions are fully positive (Pesonen 1978, 1979).

(2) A fully positive baked contact test at site 3B demonstrates that the Copper Harbor lava flow there has not been reset. Furthermore, its direction of magnetization is very similar to that of the underlying Portage Lake volcanics, which themselves show no sign of being significantly contaminated by a secondary component (Books 1972). On geological grounds White (1972) has argued that the Copper Harbor formation is more closely related temporally with the Portage Lake sequence than with younger sedimentary formations, so that a similarity in paleomagnetic direction might be expected.

(3) Remagnetization circle data suggest that the overprint in the Mamainse Point conglomerate has a direction that is very similar to overlying N polarity volcanics and thus cannot be responsible for the repeated asymmetry found in this area (Palmer 1970; Robertson 1973).

We may therefore conclude that remagnetization in Keweenawan igneous units is *not* responsible for observed asymmetry and other explanations such as those shown in Fig. 2 must be sought.

At the present time much of the polar distribution shown in Fig. 1 appears to be consistent with APW. This is particularly evident in the uppermost N polarity group dominated by units of known stratigraphic order from the Keweenaw Peninsula (poles PL, NO, and FR in Fig. 1). The same counterclockwise sense of movement is also evident in the older R polarity group, although here stratigraphic control is less secure (Halls and Pesonen 1981). In most places around Lake Superior where a polarity reversal occurs, the sense of polar movement within the N and R polarity groups is the same as that of the R → N displacement, which may thus suggest that the reversal asymmetry is also due to APW. However, recent geochemical observations by Massey (1979) suggest that the three reversals found at Mamainse Point are not the result of fault duplication, and therefore that the repeated reversal asymmetry is a real stratigraphic feature. Since remagnetization no longer provides a valid explanation for this asymmetry, and APW demands an oscillating polar movement relative to the continent that is in phase with polar reversals, the only recourse is to propose that the asymmetry is due to some nondipole behaviour such as outlined in models 3 and 4 of Fig. 2.

If three reversals *are* present during the time of Keweenawan igneous activity it may be premature to correlate Keweenawan volcanic sequences on the as-

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sumption of a single R → N polarity change, as originally cautioned by Craddock (1972), and also to use the reversal data as a basis for time stratigraphic subdivisions of Keweenawan rocks (e.g., Books 1968; Green 1977).

One final aspect concerns the nature and extent of the overprints found in the Mamainse and Copper Harbor conglomerates. Site 4, in the Great Conglomerate at Mamainse Point, lies about 375 m stratigraphically below a 2000 m thick sequence of N polarity volcanics. The Great Conglomerate would thus have been subjected to increased temperature and chemical activity during this major volcanic episode. The similarity in magnetization direction between the volcanics and the conglomerate overprint suggests that the latter was indeed formed in such a thermochemical regime.

This interpretation cannot hold for the Copper Harbor overprint, not only because of a complete absence of overlying flows at some sites, but also because the overprint direction is very different from that of primary remanences found in either the Portage Lake or Copper Harbor volcanics. The polar distribution in Fig. 1 and evidence from Copper Harbor flows (Halls and Palmer 1981) suggest that the overprint postdates the entire Copper Harbor sequence and is more closely correlative in time with the overlying Freda and Nonesuch formations. However, these units are conformable with the Copper Harbor and the overprint predates tilting so that the remagnetization event must be younger than or perhaps equivalent to the uppermost parts of the Freda.

One of the assumptions inherent in Palmer's (1970) remagnetization hypothesis is that a degree of resetting similar to or greater than that exhibited by Keweenawan conglomerates also occurs in older volcanic and intrusive units. This assumption would be reasonable had the overprint been of PTRM nature due, for example, to a high geothermal gradient. However, our demonstration that the overprint is of a CRM origin no longer demands that older units are likewise affected, because CRM acquisition may be mostly a function of fluid permeability.

A possible explanation, therefore, for the apparent lack of the Copper Harbor overprint in the underlying Portage Lake volcanics is that the conglomerate was more permeable and also that the lava pebbles, having been previously weathered during transport, were more susceptible to chemical attack. Within the Portage Lake volcanics secondary mineral infilling (including native copper) is concentrated in amygdaloidal and rubbly flow tops and it can be shown that these once permeable rocks, rather than the more massive flow interiors, provided channelways for solutions. Further work is obviously needed to paleomagnetically compare different parts of the individual Portage Lake flows to see whether some of the mineralization may date from the time of the Copper Harbor overprint.

Acknowledgements

The authors would like to thank K. G. Books, formerly with the U.S. Geological Survey, Reston, Virginia for unpublished NRM intensity data on the Powdermill Group, Dagmar Lacina for assistance with some of the remanence measurements in the UWO laboratory, and J. C. Briden for constructive criticism of the original manuscript. Suggestions and constructive criticisms by the reviewers, C. S. Grommé and W. A. Morris, improved the final manuscript. W. A. Morris suggested the modal analysis and calculated the modes for us using a computer program supplied to him by D. R. Van Alstine. Financial support for this project has been obtained through Natural Sciences and Engineering Research Council of Canada grants A7824 (HCH and LJP) and A3554 (HCP).

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